

PRODUCT BACKLOG

CeDInt-SEF | DataPack

Client / Product View

Smart Energy Forecasting for the CeDInt Building at UPM

Scrum Product Backlog - English Version

PM² governance + Scrum/Agile execution + CRISP-DM

Project	CeDInt-SEF: Smart Energy Forecasting / DataPack
Product	Analytical and predictive DataPack for CeDInt energy consumption
Perspective	Client / stakeholder needs and product capabilities
Scope basis	Approved Business Case and existing project backlog

1. Purpose of this Product Backlog

This Product Backlog describes the product from the client's perspective. Each item answers the question: “As a client, what do I want the product to enable me to do, or what do I want to receive from the project?”

Technical implementation activities are intentionally excluded from the Product Backlog. Tasks such as coding a resampling function, detecting device resets, cleaning individual files, testing a model or producing a graph are implementation tasks. They belong in the Sprint Backlog after a Product Backlog Item has been selected.

The backlog is grounded in the project scope: historical IoT/Excel extracts, Building → Floor → Zone spatial context, energy-consumption forecasting, data-quality transparency, model evaluation, communication of results and reproducible handover.

2. Product Vision

For UPM stakeholders and energy/building analysts who need to understand and anticipate CeDInt's energy consumption, DataPack is an analytical and predictive solution that transforms historical sensor data into understandable, spatially contextualised and evaluated energy insights and forecasts.

The product is successful when a client can understand what data is available, judge its reliability, analyse consumption in its physical context, review energy forecasts and their performance, understand the main findings and receive a reproducible final package.

3. Product Users / Stakeholders

User / stakeholder	Main need	Expected product value
UPM / Project Sponsor	Assess whether the project produces credible evidence.	Clear results, performance, limitations and final package.
Energy / Building Analyst	Understand past consumption and what may happen next.	Historical analysis, spatial context and forecasts.
Data / ML Reviewer	Assess whether the analytical results are trustworthy.	Transparent data quality, modelling evidence and evaluation.
Professor / Evaluator	Verify that the project follows a sound methodology.	Traceable requirements, justified results and reproducible evidence.

4. Client User Journeys

Journey A - Understand the building data

1. Select a building area and period.
2. See which data and measurements are available.
3. Understand what the measurements represent.
4. Assess data quality and known limitations.
5. Decide whether the data is suitable for analysis.

Journey B - Analyse energy consumption

1. Select a relevant period and spatial scope.
2. Review historical consumption.
3. Compare consumption across available zones.
4. Identify relevant temporal or spatial patterns.
5. Use the evidence to understand energy behaviour.

Journey C - Obtain and assess an energy forecast

1. Define the prediction case and forecast horizon.
2. Obtain an energy-consumption forecast.
3. Compare forecast performance with an appropriate reference.
4. Review predicted versus observed values where available.
5. Understand model performance and limitations.

Journey D - Review and reuse the result

1. Open the final DataPack.
2. Understand the methodology and key decisions.
3. Review the evidence behind the results.
4. Access the reproducible analysis package.
5. Record acceptance, limitations and recommendations.

5. Client-Oriented Product Backlog

Priority uses MoSCoW: Must = required for the core product; Should = important but deferrable; Could = useful enhancement if time and evidence permit.

ID	Product capability	Client User Story	Priority
PB-01	Explore available energy data	As a client, I want to see which energy data sources, variables, time periods and sampling characteristics are available so that I can understand the analytical scope.	Must
PB-02	Understand measurements	As a client, I want to understand the meaning, units and context of the available measurements and device families so that I can correctly interpret the energy data.	Must
PB-03	Assess data quality	As a client, I want to see the main data-quality characteristics and known limitations so that I can judge whether the available data is reliable enough for analysis.	Must
PB-04	Access a trusted analytical dataset	As a client, I want to receive a controlled and documented analytical dataset so that the energy analysis and forecasting are based on data whose preparation is traceable.	Must
PB-05	Explore energy in spatial context	As a client, I want to relate energy measurements to the Building → Floor → Zone structure so that I can understand where consumption occurs.	Must
PB-06	Analyse historical consumption	As a client, I want to analyse historical energy consumption over time and across available zones so that I can identify relevant patterns and behaviours.	Must
PB-07	Use contextual information	As a client, I want energy analysis and forecasts to consider relevant temporal, environmental and calendar context when supported by the data so that the results are better informed.	Should
PB-08	Obtain energy forecasts	As a client, I want to obtain forecasts of energy consumption for a defined future period so that I can anticipate expected consumption.	Must
PB-09	Compare forecasting approaches	As a client, I want to compare the performance of the forecasting approaches tested so that I can identify the most appropriate candidate for the project.	Must
PB-10	Assess forecast reliability	As a client, I want to see forecast performance, predicted-versus-observed results and limitations so that I can understand how much confidence I can place in the predictions.	Must
PB-11	Identify energy insights	As a client, I want clear evidence-based insights about consumption patterns and relevant spatial or temporal behaviours so that I can understand what the data reveals.	Must
PB-12	Review results clearly	As a client, I want the main analyses, forecasts, performance results and limitations to be presented clearly so that technical and non-technical stakeholders can understand the outcome.	Must
PB-13	Reuse and reproduce the analysis	As a client, I want the final analysis, methodology, data dictionary and model documentation to be organised and reproducible so that the work	Must

		can be reviewed and reused.	
PB-14	Receive the final DataPack	As a client, I want to receive a complete final DataPack containing the agreed results, documentation, code and supporting evidence so that the project can be formally reviewed and handed over.	Must

6. Acceptance Criteria

ID	Acceptance criteria
PB-01	The available sources, variables, relevant time coverage and sampling behaviour can be reviewed; the scope of available data is clear.
PB-02	Key measurements have an agreed interpretation and units; device/telemetry terminology is documented sufficiently for a user to understand the data.
PB-03	The client can identify the principal missingness, irregular-sampling, anomaly and other known data-quality limitations; unresolved issues are visible.
PB-04	A versioned analytical dataset is available; its preparation is traceable; known data-quality treatments are documented; raw source data is not silently overwritten.
PB-05	Available records can be interpreted through Building → Floor → Zone where source evidence supports the mapping; unmapped or ambiguous cases are visible.
PB-06	Historical consumption can be reviewed by time and available spatial scope; the resulting analysis supports identification of meaningful patterns.
PB-07	Relevant environmental, temporal or calendar information is included only where supported by the available data and justified by the analysis.
PB-08	A functional energy-consumption forecast can be generated for a defined prediction case and horizon; the output is understandable to the intended user.
PB-09	At least one baseline/reference and candidate forecasting approach can be compared using a documented evaluation protocol.
PB-10	Forecast performance is reported using appropriate metrics and later/unseen time periods; predicted and observed values can be compared where available; limitations are stated.
PB-11	The final product contains evidence-based energy insights, with spatial and temporal interpretation where supported by the data.
PB-12	The final communication presents the main findings, forecast performance and limitations in a form suitable for stakeholder review and presentation.
PB-13	The final package contains the methodology, data dictionary, relevant analysis/model documentation and instructions needed to understand and reproduce the workflow.
PB-14	The agreed final code, documentation, results and supporting evidence are packaged for handover; excluded hardware, live-pipeline modification and real-time streaming are not part of the delivered product.

7. Product Backlog vs. Sprint Backlog

The following distinction is essential for the Scrum implementation:

Product Backlog - WHAT the client wants	Sprint Backlog - HOW the team builds it
"I want to assess the quality of the available data."	Profile missing intervals, sampling gaps, anomalies and quality flags.
"I want a trusted analytical dataset."	Define resampling rules, treat resets, standardise timestamps and generate the dataset.
"I want to analyse consumption by zone."	Map device IDs to zones, join datasets and create analysis tables/plots.
"I want an energy forecast."	Create features, split data temporally, train models and calculate metrics.
"I want to compare forecasting approaches."	Train candidate models and run the evaluation protocol.
"I want a clear final result."	Build figures, tables, narrative and presentation material.

8. Suggested Product Increments

Increment	Client-visible outcome	Relevant PBIs
Increment 1 - Data can be understood	The client can see what data exists, what it means and its principal quality limitations.	PB-01 to PB-03
Increment 2 - Data can be trusted and located	A controlled analytical dataset and spatial context are available for analysis.	PB-04 to PB-05
Increment 3 - Consumption can be analysed	Historical energy behaviour and relevant context can be explored.	PB-06 to PB-07
Increment 4 - Forecast can be assessed	Energy forecasts, model comparisons and reliability evidence are available.	PB-08 to PB-10
Increment 5 - Results can be communicated and reused	Insights, final communication and a reproducible DataPack are delivered.	PB-11 to PB-14

CeDInt-SEF / DataPack - Client-Oriented Product Backlog